WATER POLLUTION & DISEASE

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DATASET

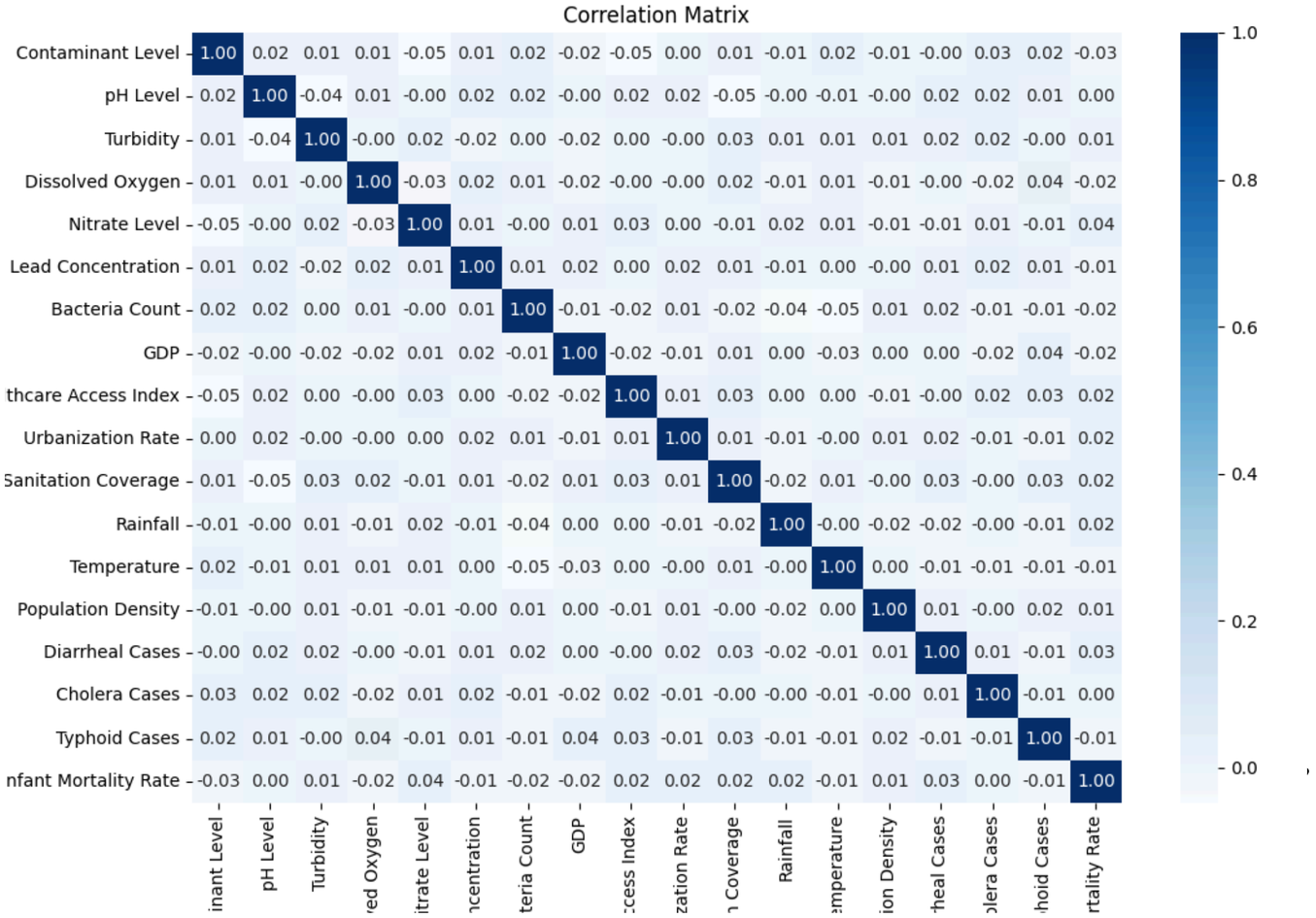
This dataset explores the relationship between water pollution and the prevalence of waterborne diseases worldwide.
It includes water quality indicators, pollution levels, disease rates, and socio-economic factors that influence health outcomes.
The dataset provides information on different countries and regions, spanning the years 2000-2025.
It covers key factors such as contaminant levels, access to clean water, bacterial presence, water treatment methods, sanitation coverage, and the incidence of diseases like diarrhea, cholera, and typhoid. Additionally, it incorporates socio-economic variables such as GDP per capita, urbanization rate, and healthcare access, which help assess the broader impact of water pollution on communities.

Prevalence: Covers 10 countries (e.g., USA, India, China, Brazil, Nigeria, Bangladesh, Mexico, Indonesia, Pakistan, Ethiopia).
Includes 5 regions per country (e.g., North, South, East, West, Central).
Spans 26 years (2000-2025).
Features 3,000 unique records representing various water sources and pollution conditions.

HEATMAP

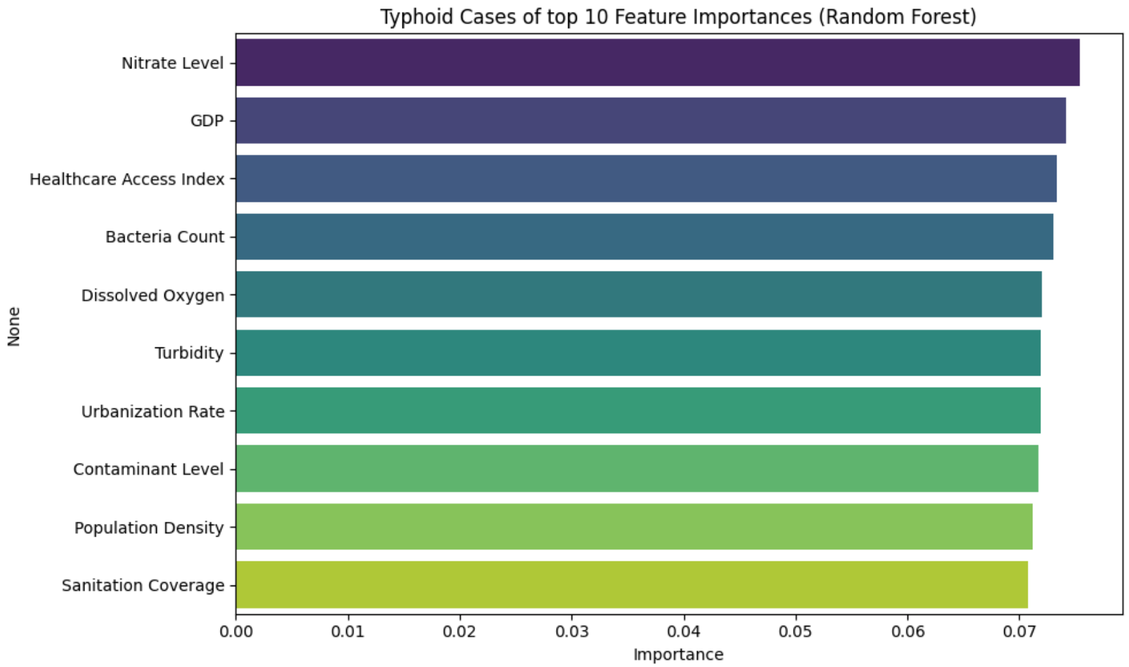
We can see from the heat map that all water quality factors and disease incidence rates have low correlation with each other.

The total amount of data is 24 columns*3000 rows
The factors in this figure do not include
Country、Region、Year
Water Source Type
Water Treatment Method
Access to Clean Water

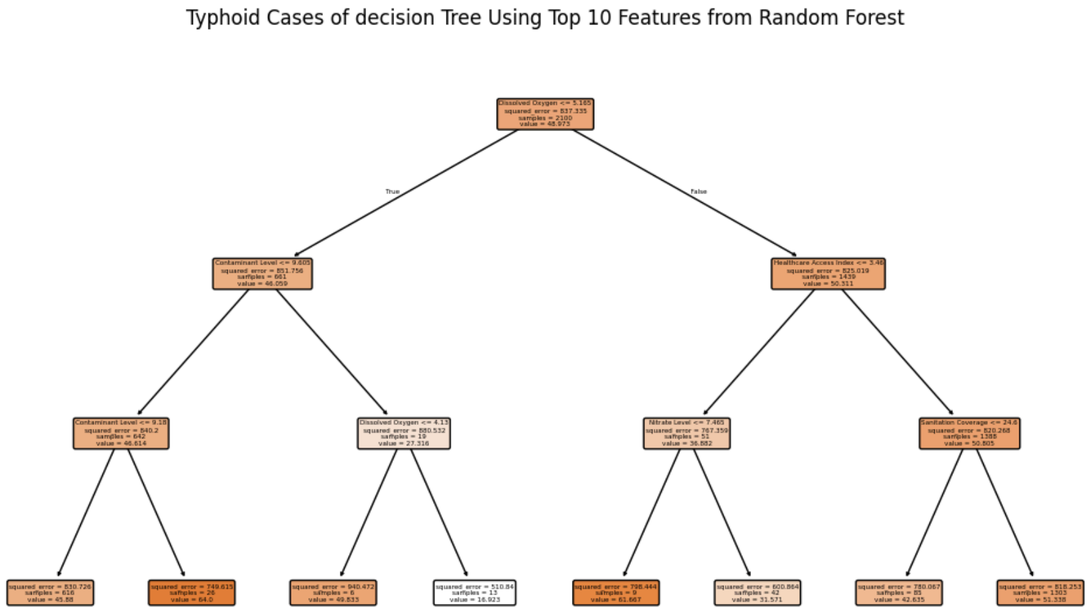


TYPHOID CASES

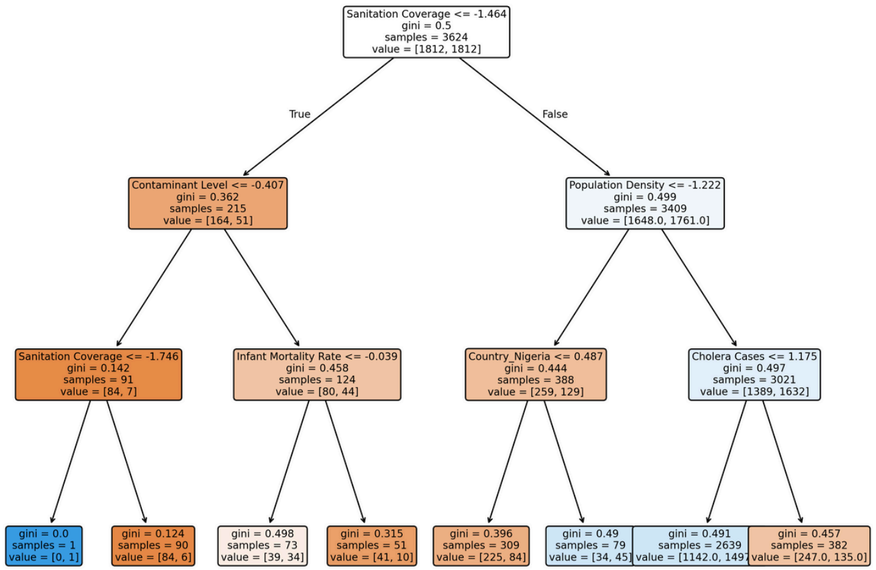
Typhoid and paratyphoid are both intestinal infectious diseases caused by bacteria, the pathogens are Salmonella typhi and Salmonella paratyphi respectively.
Typhoid and paratyphoid are global diseases, with cases occurring all over the world, but most of them occur in developing countries, especially in areas where tap water is not widely available or environmental sanitation is poor.
The risk of infection is highest in South Asia, and there are also higher risks in Southeast Asia, Africa, Central and South America, and the Caribbean Islands.



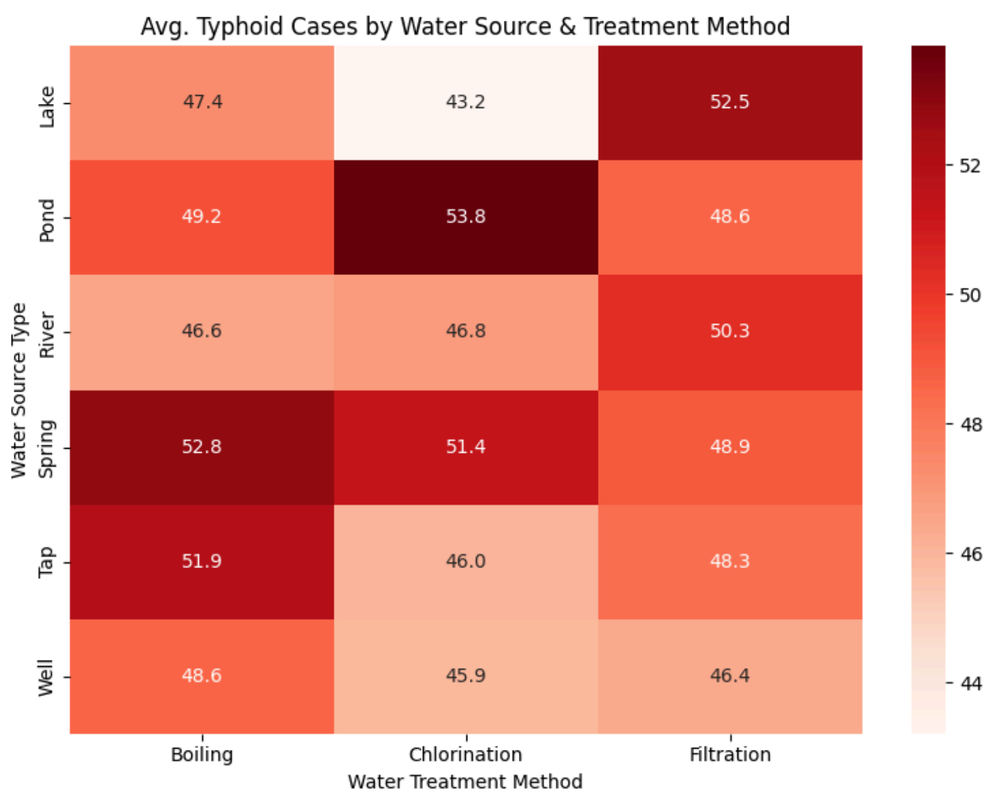
- Top 10 feature importance for random forest model



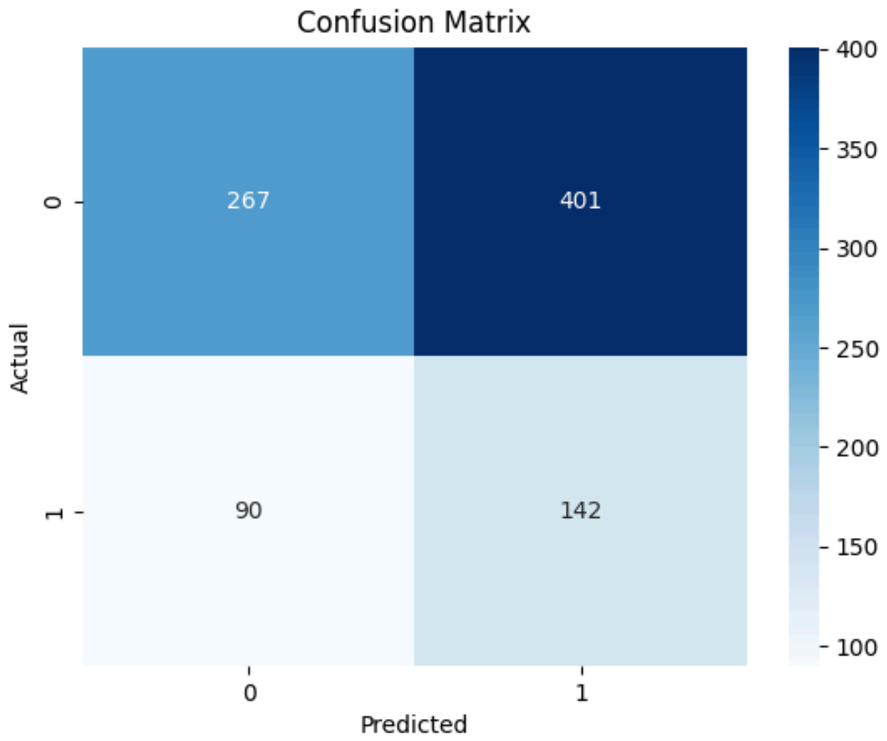
- Decision Tree Prediction Model
(First 10 Features from Random Forest)



TYPHOID CASES& WATER SOURCE TYPE, WATER TREATMENT METHOD

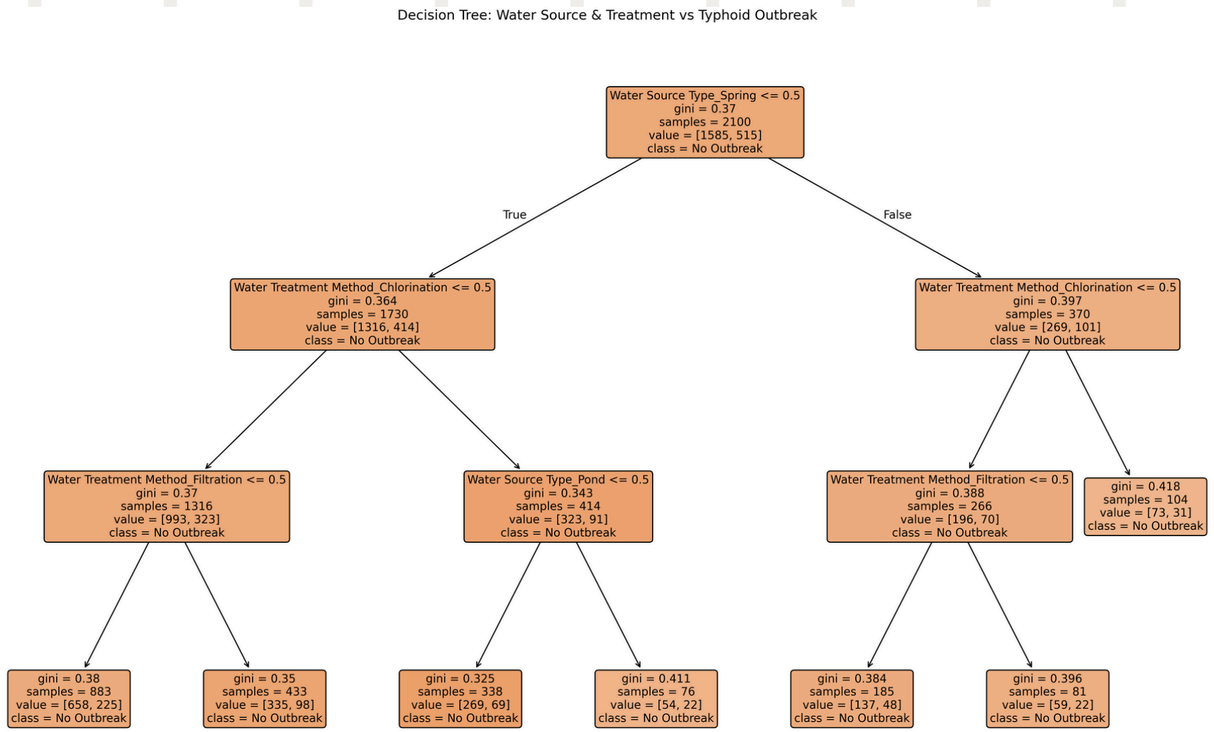


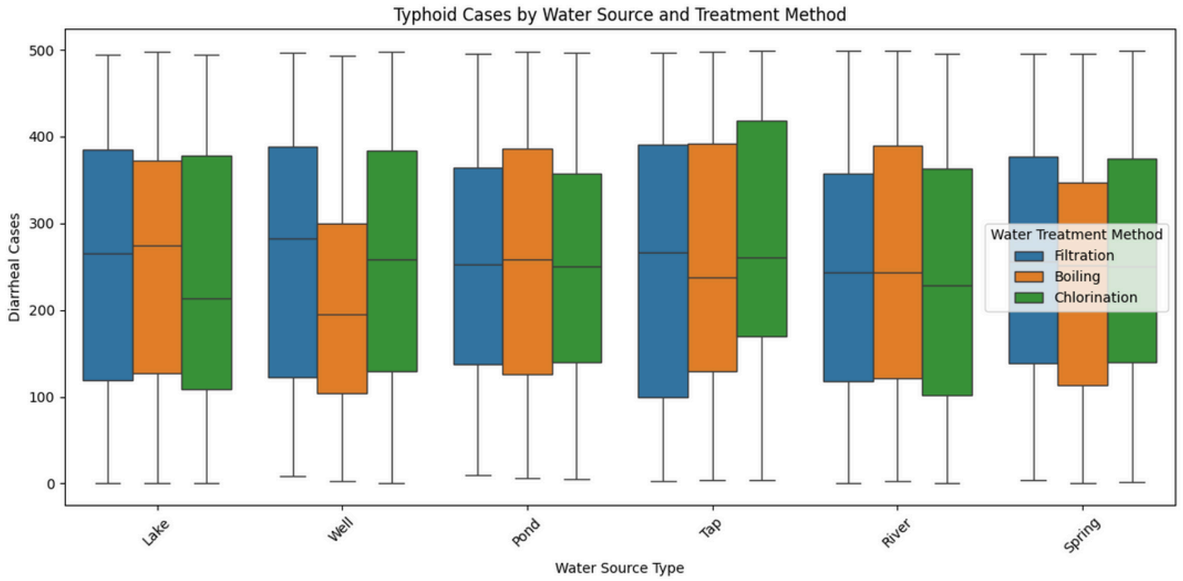
- Heatmap of Typhoid Cases by Water Source & Treatment Method



- Confusion Matrix of Typhoid Cases

TYPHOID CASES& WATER SOURCE TYPE, WATER TREATMENT METHOD



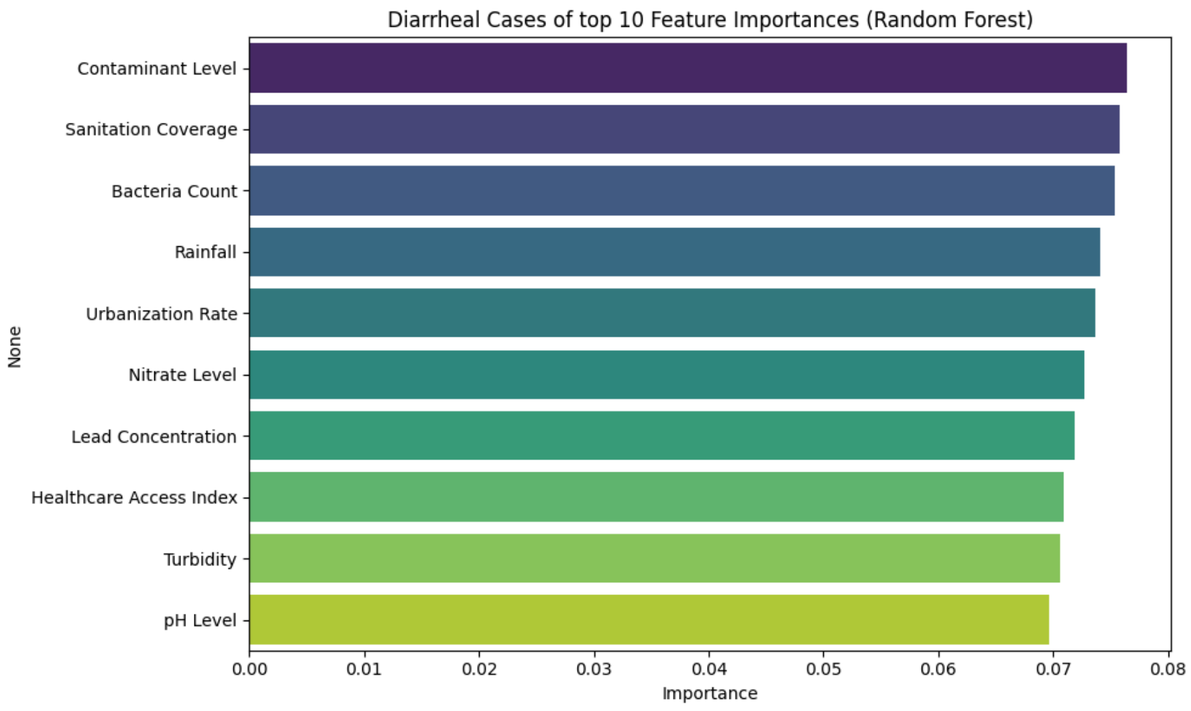


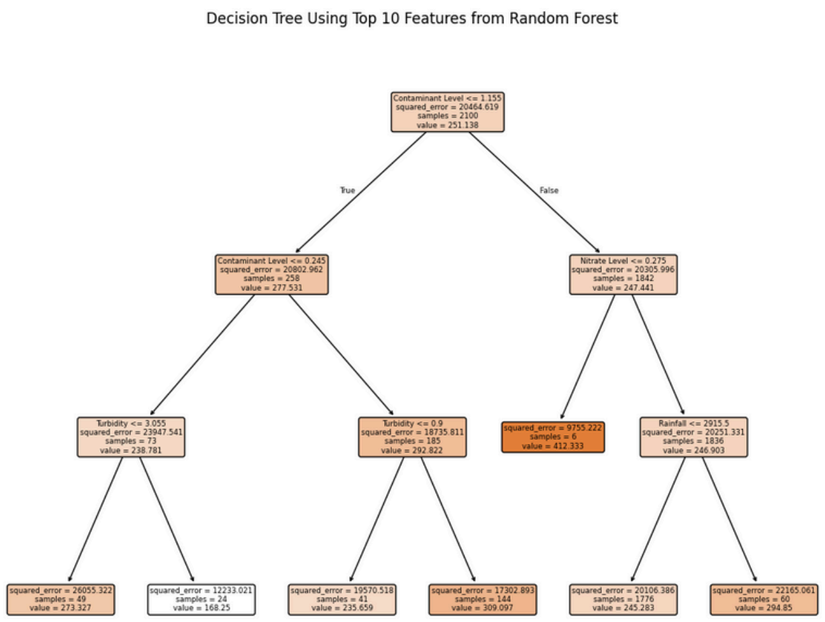
- We can see that the median of the combination of "Tap + Chlorination" is slightly higher, while the distribution of "Well + Filtration" is more concentrated.

- The model ultimately predicts most areas as “not outbreak-free,” but high-risk combinations (such as Pond + Untreated + No Chlorine) can also be observed from each terminal node.

1. Comprehensive analysis results show that the occurrence of typhoid fever is closely related to "water source selection", "treatment method" and "pollution level".
2. Boiling treatment is the best means of epidemic prevention under most conditions; pollutants (such as lead, nitrates) and increased bacteria in water will significantly increase the risk of epidemics.
3. Although the model cannot accurately predict the outbreak of an epidemic, it can help us identify high-risk combinations and areas, and then design targeted prevention strategies.

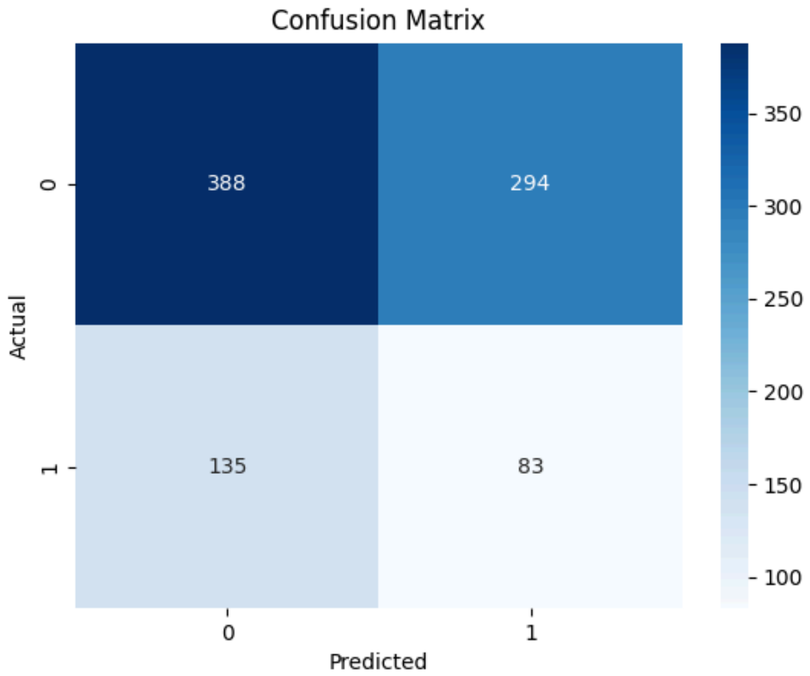
DIARRHEAL CASES



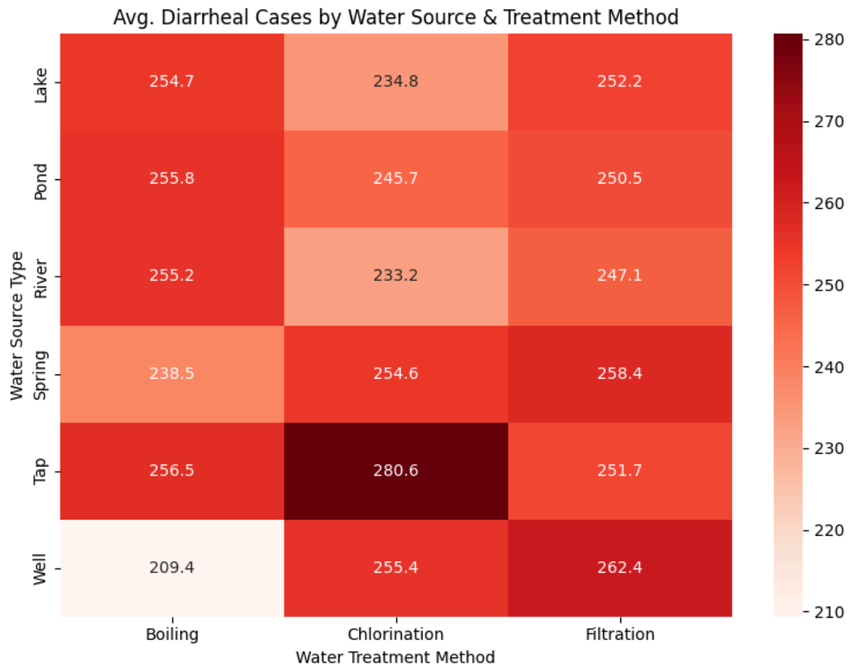


- Understand how environmental factors affect the severity of the epidemic layer by layer

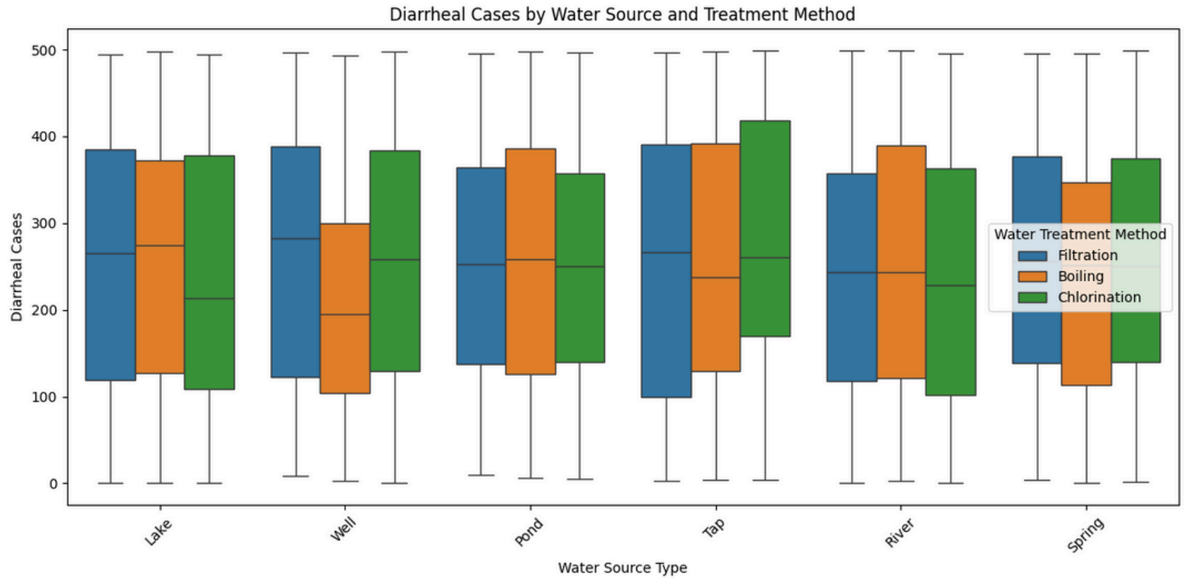
DIARRHEAL CASES & WATER SOURCE TYPE, WATER TREATMENT METHOD



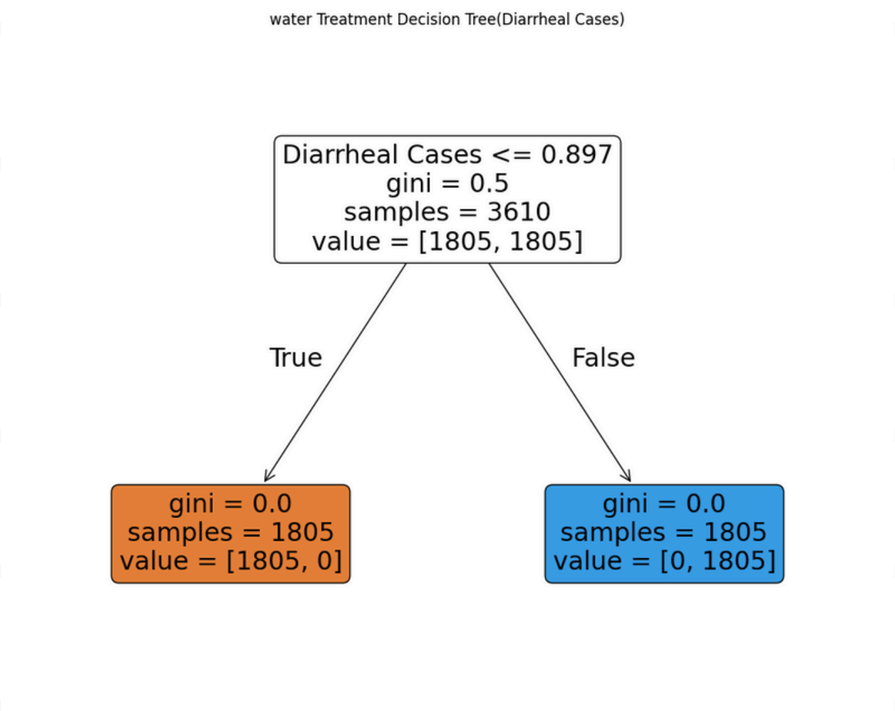
- There may be false positives, but it is not easy to miss the risk of an outbreak

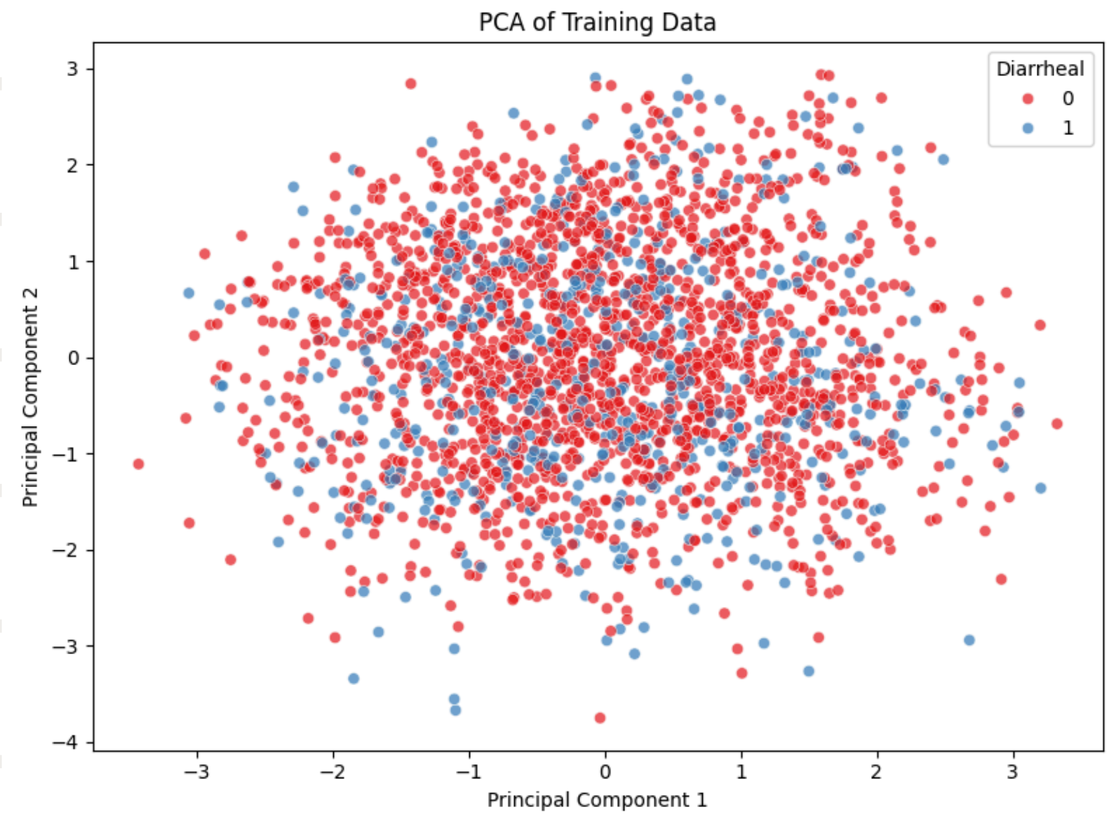


- Tap + Chlorination (tap water + chlorine treatment) reached the highest value (280.6), indicating that this combination may be seriously polluted or poorly treated; while "Well + Boiling (well water + boiling)" had the lowest number of cases (209.4), reflecting that it may be one of the safest options



- Even with the same treatment method, different water sources will affect the median and extreme values of the epidemic.





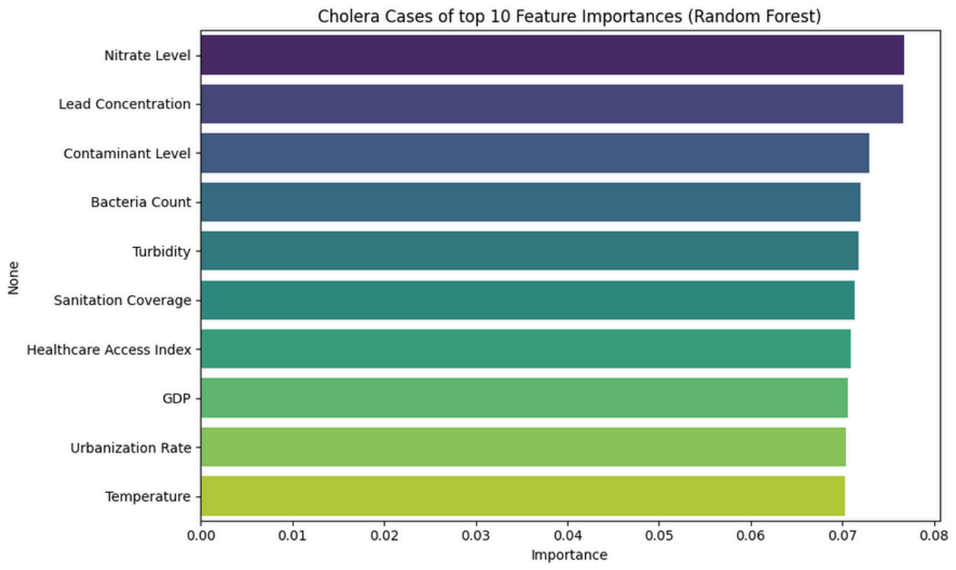
1. The data itself is difficult to classify: PCA and confusion matrix both show that the model is difficult to accurately predict outbreaks;
2. Water source and treatment methods are still influential: heat maps and box plots reveal that certain combinations (such as Tap + Chlorination) are riskier;
3. Although the decision tree is accurate, we need to be wary of overfitting: it is recommended to do cross-regional verification or external testing before actual application.

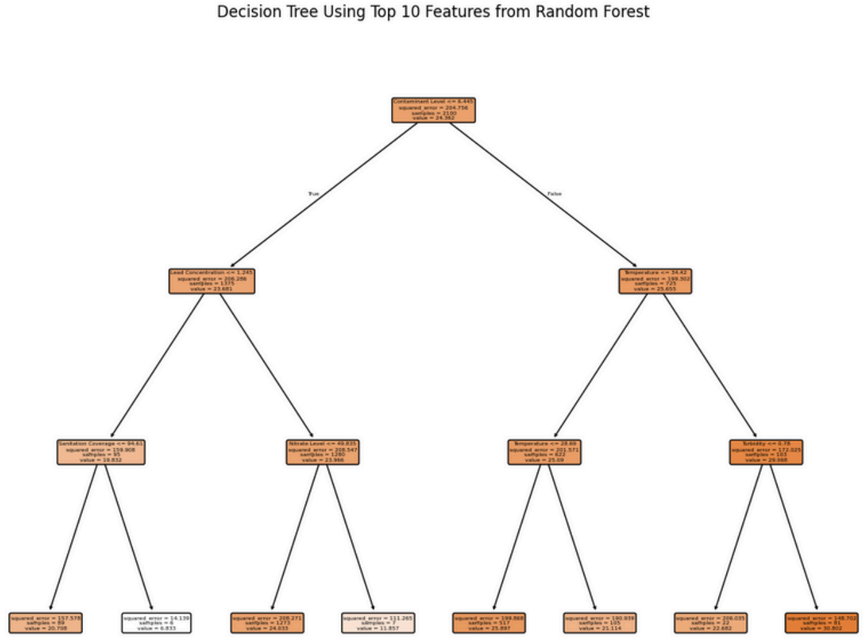
CHOLERA CASES

Acute bacterial enteric infectious disease is a global disease with cases occurring all over the world, most of which occur in developing countries, especially in areas where tap water is not widely available or where environmental sanitation is poor.

Symptoms of cholera infection include painless, massive, watery diarrhea, vomiting, and rapid dehydration.

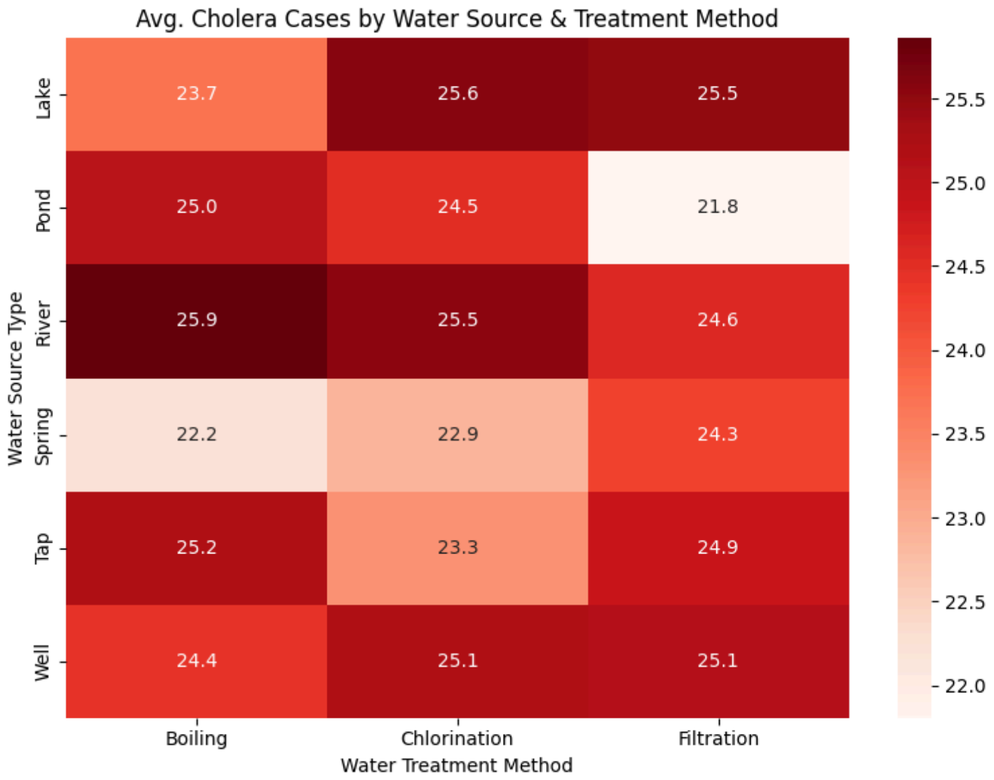
If the patient fails to receive appropriate treatment in time, he or she may die from severe dehydration and shock.



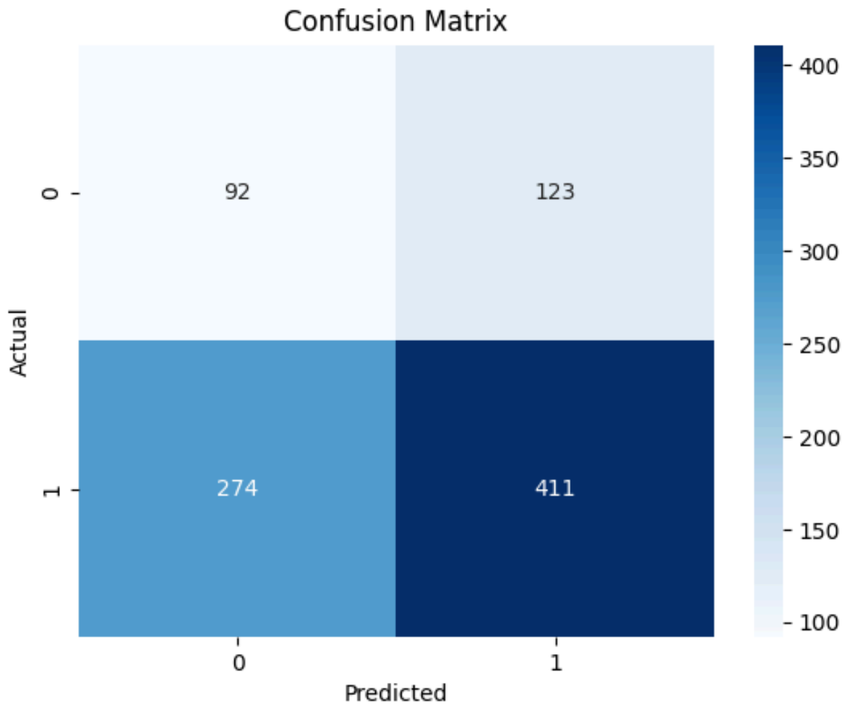


- Regression decision tree generated by the top ten important features in random forest

CHOLERA CASES &WATER SOURCE TYPE, WATER TREATMENT METHOD

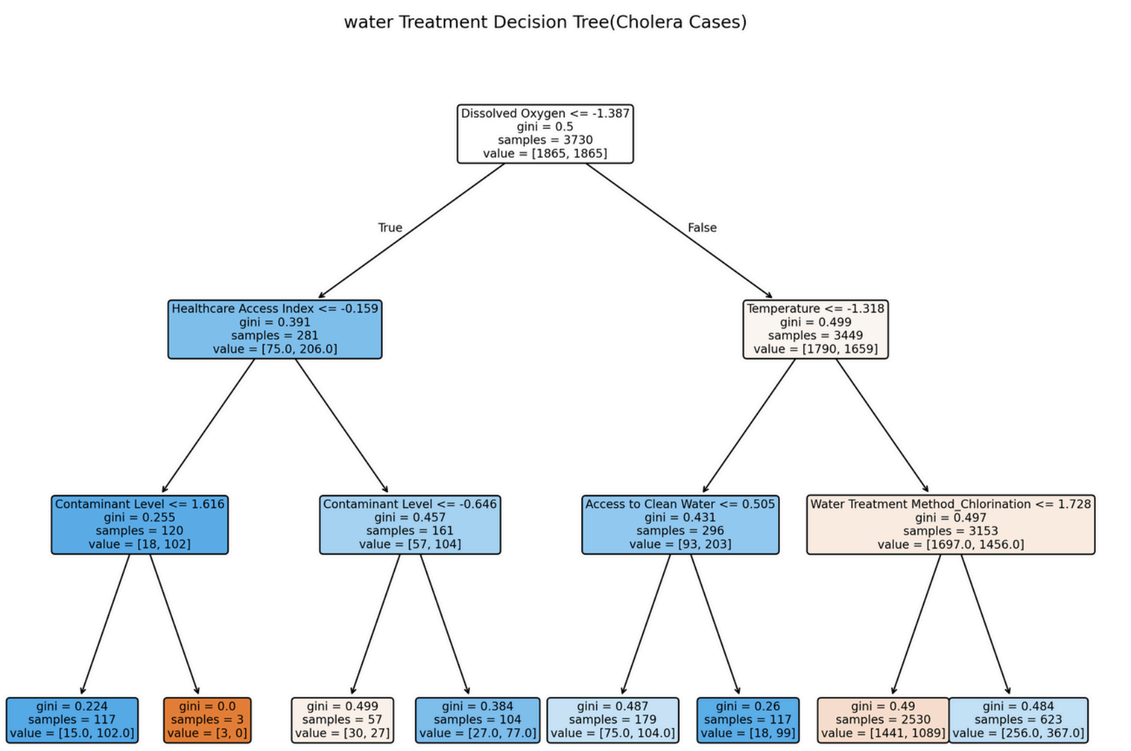


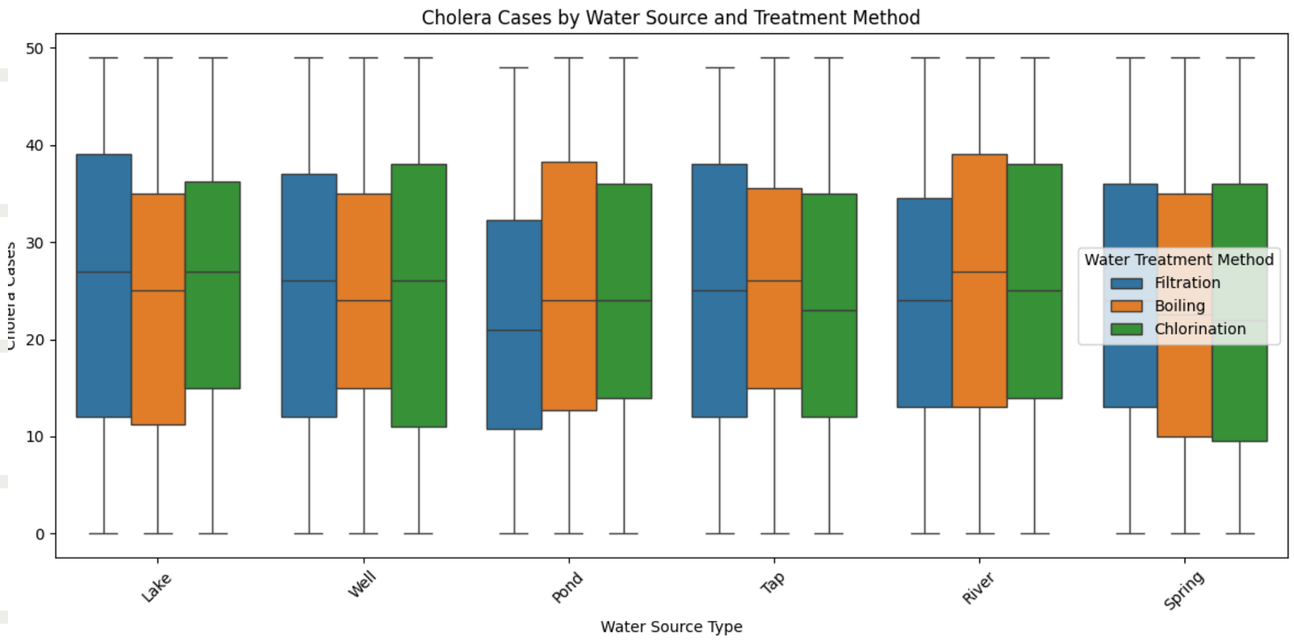
- Each cell in the heat map represents the average number of cholera cases under a certain water source and treatment method.
- River + Boiling is the highest (25.9), and Spring + Boiling is the lowest (22.2).
- Overall, Boiling and Chlorination perform well under most water sources, but there are still large individual differences.



- The upper left column shows the truly confirmed cases without outbreaks (TN=92), and the lower right column shows the correctly predicted outbreaks (TP=411). However, there were 274 cases that were actually outbreaks but were predicted incorrectly (FN), reflecting insufficient recall.

CHOLERA CASES&WATER SOURCE TYPE, WATER TREATMENT METHOD





1. Water sources and treatment methods significantly affect cholera risk, and high-risk combinations can be quickly identified through heat maps and box plots;
2. Pollutant concentrations (Nitrate, Lead) are strongly correlated with cholera and can be used as early warning indicators;
3. Although the classification model still has errors in outbreak prediction, the visual decision tree supplements the needs of policy understanding and risk classification.

REFERENCES

- https://www.kaggle.com/datasets/khushikyad001/water-pollution-and-disease/data
- https://www.cdc.gov.tw/Disease/SubIndex/wp-kiiztSq_5cFd6H96f9w
- https://www.cdc.gov.tw/Disease/SubIndex/FNpFzvcM-QKKt6XBHJkyxA